BIOLOGICAL SYSTEMS MODELING AND ANALYSIS

Homework Assignment 2

【Spring Semester, 2025】

Due Date: March 26, 2026

Problem 1: (25%)

The classical Lotka-Volterra predator-prey model is:

Prey: $V_{t+1} = V_t + rV_t - aV_tP_t$

Predator: $P_{t+1} = P_t + abV_tP_t - dP_t$

Assuming the units are a conserved quantity (e.g., g C), draw the Forrester diagram. The parameters are defined as: r = prey per capita rate of increase, a = rate of consumption of prey by predator, b = conversion of prey consumed to new predators, and d = predator death rate.

Problem 2: (25%)

Draw a Forrester diagram of carbon and water dynamics in a tree over a four month growing season, when the time step is one hour. The geographical setting is in the mid-latitudes, so that basic atmospheric conditions (e.g., photoperiod, light intensity, precipitation) change significantly during the growing season. The basic relationships are as follows.

In the roots, water and oxygen are taken-up, sugars manufactured in the leaves are used for cellular respiration, and CO_2 is released as a by-product. Water is transported upwards to leaves within the xylem where it increases the rate of uptake of atmospheric CO_2 (via stomata). CO_2, H_2O, and light combine to produce, among other essential molecules, sugars that are used in the leaves and transported downward within the phloem for use by the roots. When water level in the leaves decreases to a low level, the stomata close to reduce water loss (transpiration), little CO_2 enters the leaves and photosynthesis and the rate of sugar production decreases. When leaf water level is high, the water loss rate is high, but the rate of CO_2 entering the leaves is also high, consequently increasing photosynthesis rate.

Problem 3: (25%)

A simple foodweb (1 prey, 2 predators) is modeled in units of grams of carbon. A prey species (x) grows according to density-dependent growth in the absence of either predator. The consumption of x per unit of predator 1 (y) is a saturation feedback function and the consumption of x by predator 2 (z) is a fixed fraction of x. The per capita death rate of y is constant. The per capita death rate of z is a negative exponential. Draw a Forrester diagram and write differential equations with the above hypotheses. (Consult Fig. 5.4 if you need help with some of the functional forms alluded to.)

Problem 4: (25%)

Write four differential equations for this scenario. In an animal's immune system, suppose there is a population of cancerous cells (C) that kill healthy (H) cells. The kill rate is proportional to the mass action between C and H. Without cancer, the healthy cells grow according to self-inhibition to reach a constant value T. C cells are killed by two forms of white blood cells: M and K. These kill C cells by a mass action process, but both M and K are required for a successful kill. Assume, the two white blood cells move randomly and that they divide at a rate that is proportional to the number of cancer cells in the system. In the absence of cancer, the white blood cells decline exponentially to a small, non-zero level and remain at that level until more C cells are present.

Verify that the units of your model are correct.

Notes:

1. Please submit your homework report together with your program to the NTU COOL course website.
2. Late submission will have a penalty of 10% discount per day of your homework total score toward a maximum of 50% discount. No late submission over five days will be accepted.